

# REPORT

## Title

## Prolog Program to Implement Towers of Hanoi

## Aim

To implement the **Tower of Hanoi problem using recursion in Prolog.**

## Problem Statement

Implement a Prolog program that moves a given number of disks from one peg to another using an auxiliary peg while following the rules of the Tower of Hanoi.

## Requirements

- SWI-Prolog
- Computer system
- Prolog source file (`exp5.pl`)

## Procedure

1. Open **SWI-Prolog**.
2. Create a new file named `exp5.pl`.
3. Enter the Tower of Hanoi Prolog program.
4. Save the file.
5. Load the program using `[exp5]..`
6. Execute the query `hanoi(3, left, right, center) ..`
7. Observe the sequence of disk movements.

## Program

```
hanoi(1, Source, Destination, _) :-  
    write('Move disk from '),  
    write(Source),  
    write(' to '),  
    write(Destination),  
    nl.  
  
hanoi(N, Source, Destination, Auxiliary) :-  
    N > 1,  
    N1 is N - 1,  
    hanoi(N1, Source, Auxiliary, Destination),  
    write('Move disk from '),  
    write(Source),  
    write(' to '),  
    write(Destination),  
    nl,  
    hanoi(N1, Auxiliary, Destination, Source).
```

## Sample Output

### Query:

```
?- [exp5].  
true.
```

```
?- hanoi(3, left, right, center).
```

### Output:

```
Move disk from left to right  
Move disk from left to center  
Move disk from right to center  
Move disk from left to right
```

```
Move disk from center to left  
Move disk from center to right  
Move disk from left to right  
true.
```

## **Conclusion**

Thus, the **Tower of Hanoi problem** was successfully implemented in Prolog **using recursion**. For 3 disks, the program performs the required **7 moves** to transfer all disks from the source peg to the destination peg.